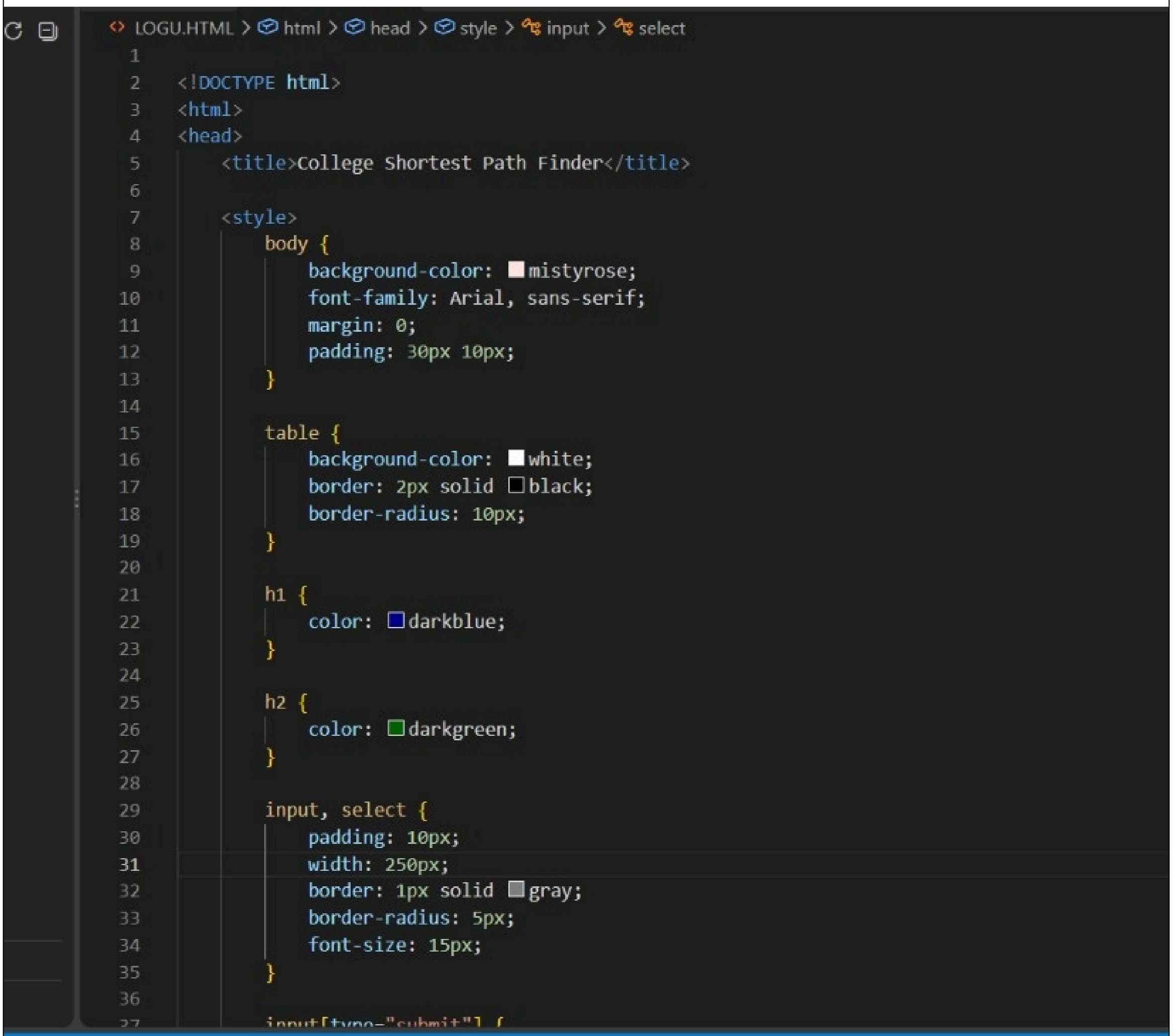

LOGU.HTML > html > head > style > input > select
<!DOCTYPE html>
<html>
<head>
<title>College Shortest Path Finder</title>
<style>
body {
background-color: mistyrose;
font-family: Arial, sans-serif;
margin: 0;
padding: 30px 10px;
}
table {
background-color: white;
border: 2px solid black;
border-radius: 10px;
}
h1 {
color: darkblue;
}
h2 {
color: darkgreen;
}
input, select {
padding: 10px;
width: 250px;
border: 1px solid gray;
border-radius: 5px;
font-size: 15px;
}

```
<html>
<head>
    <style>
        }

        #result {
            color: darkgreen;
            font-weight: bold;
            line-height: 1.8;
            overflow-wrap: anywhere;
        }

        .place {
            font-size: 38px;
            padding: 8px;
        }

        .caption {
            font-size: 13px;
            color: #444;
        }

        .note {
            font-size: 13px;
            color: gray;
        }
    </style>

    <script>
        const places = [
            "Main Gate",
            "Library",
            "Canteen",
            "CSE Block",
            "Laboratory",
            "Auditorium"
        ];
```

LOGU.HTML > html > head > style > input > select

```
 3   <html>
 4   <head>
72       <script>
73           const places = [
80           ];
81
82           const INF = Infinity;
83
84           // Distances between locations in meters
85           const graph = [
86               [0,   100, 200, INF, INF, INF],
87               [100, 0,   50,  150, INF, INF],
88               [200, 50,  0,   100, 200, INF],
89               [INF, 150, 100, 0,   50,  120],
90               [INF, INF, 200, 50,  0,   80],
91               [INF, INF, INF, 120, 80,  0]
92           ];
93
94       function findShortestPath(event) {
95           event.preventDefault();
96
97           const source = Number(
98               document.getElementById("source").value
99           );
100
101          const destination = Number(
102              document.getElementById("destination").value
103          );
104
105          const n = places.length;
106          const dist = Array(n).fill(INF);
107          const visited = Array(n).fill(false);
108          const parent = Array(n).fill(-1);
109
110          dist[source] = 0;
111
```

LOGO.HTML > html > head > style > input > select

```
<html>
<head>
    <script>
        function findShortestPath(event) {

            // Dijkstra's algorithm
            for (let count = 0; count < n; count++) {
                let u = -1;

                for (let i = 0; i < n; i++) {
                    if (!visited[i] &&
                        (u === -1 || dist[i] < dist[u])) {
                        u = i;
                    }
                }

                if (u === -1 || dist[u] === INF) {
                    break;
                }

                visited[u] = true;

                for (let v = 0; v < n; v++) {
                    if (!visited[v] &&
                        graph[u][v] !== INF &&
                        graph[u][v] > 0 &&
                        dist[u] + graph[u][v] < dist[v]) {

                        dist[v] = dist[u] + graph[u][v];
                        parent[v] = u;
                    }
                }
            }

            const result = document.getElementById("result");

            if (dist[destination] === INF) {
```

LOGU.HTML > html > head > style > input > select

```
3    <html>
4    <head>
72       <script>
94           function findShortestPath(event) {
144                  result.textContent = "No route is available.";
145                  return;
146              }
147
148              // Reconstruct the shortest route
149              const path = [];
150              let current = destination;
151
152              while (current !== -1) {
153                  path.unshift(places[current]);
154                  current = parent[current];
155              }
156
157              result.innerHTML =
158                  "<strong>Starting Point:</strong> " +
159                  places[source] +
160                  "<br><strong>Destination:</strong> " +
161                  places[destination] +
162                  "<br><strong>Shortest Route:</strong> " +
163                  path.join(" &rarr; ") +
164                  "<br><strong>Total Distance:</strong> " +
165                  dist[destination] + " meters";
166          }
167
168          function showAllDistances() {
169              const list = document.getElementById("distanceList");
170
171              list.innerHTML = "";
172
173              for (let i = 0; i < places.length; i++) {
174                  for (let j = i + 1; j < places.length; j++) {
175                      const distance = graph[i][j];
```

LOGU.HTML > html > head > style > input > select

```
  3   <html>
  4   <head>
 72       <script>
168           function showAllDistances() {
176
177                       if (distance !== INF) {
178                           const row = document.createElement("li");
179                           row.textContent =
180                               places[i] + " to " + places[j] +
181                               " : " + distance + " meters";
182                           list.appendChild(row);
183                       }
184                   }
185               }
186           }
187       </script>
188   </head>
189
190   <body>
191
192       <form onsubmit="findShortestPath(event)">
193           <table align="center" width="750" cellpadding="12">
194
195               <tr>
196                   <td colspan="2" align="center">
197                       <h1>College Shortest Path Finder</h1>
198                       <p>Find the shortest route inside your college campus</p>
199                   </td>
200               </tr>
201
202               <tr>
203                   <td colspan="2" align="center">
204                       <span class="place">🏫</span>
205                       <span class="place">🧤</span>
206                       <span class="place">🐱</span>
207                       <span class="place">💻</span>
208                       <span class="place">🏠</span>
```

```
<body>
    <form onsubmit="findShortestPath(event)">
        <table align="center" width="750" cellpadding="12">
            <tr>
                    <span class="place">💻</span>
                    <span class="place">🔬</span>
                    <span class="place">🏛️</span>
                    <br>
                    <span class="caption">
                        Main Gate | Library | Canteen |
                        CSE Block | Laboratory | Auditorium
                    </span>
                </td>
            </tr>

            <tr>
                <td>Starting Location:</td>
                <td>
                    <select id="source" required>
                        <option value="0">Main Gate</option>
                        <option value="1">Library</option>
                        <option value="2">Canteen</option>
                        <option value="3">CSE Block</option>
                        <option value="4">Laboratory</option>
                        <option value="5">Auditorium</option>
                    </select>
                </td>
            </tr>

            <tr>
                <td>Destination Location:</td>
                <td>
                    <select id="destination" required>
                        <option value="0">Main Gate</option>
                        <option value="1">Library</option>
                        <option value="2">Canteen</option>
```

LOGO.HTML > html > body > form > table > tr > td

```
  3   <html>
190   <body>
192       <form onsubmit="findShortestPath(event)">
193           <table align="center" width="750" cellpadding="12">
232               <tr>
240                       <option value="4">Laboratory</option>
241                       <option value="5">Auditorium</option>
242                   </select>
243               </td>
244           </tr>
245
246           <tr>
247               <td colspan="2" align="center">
248                   <input type="submit"
249                          value="Find Shortest Path">
250               </td>
251           </tr>
252
253           <tr>
254               <td colspan="2" align="center">
255                   <h2>Shortest Path Result</h2>
256                   <p id="result" aria-live="polite">
257                       Select your starting location and destination.
258                   </p>
259               </td>
260           </tr>
261
262           <tr>
263               <td colspan="2" align="center">
264                   <h2>Campus Connections</h2>
265                   <p>Distances between directly connected places</p>
266                   <ul id="distanceList" style="text-align:left"></ul>
267                   <button type="button"
268                           onclick="showAllDistances()">
269                       Show Campus Distances
270                   </button>
```

LOGU.HTML > html > body > form > table > tr > td

```
  3   <html>
190   <body>
192       <form onsubmit="findShortestPath(event)">
193           <table align="center" width="750" cellpadding="12">
262               <tr>
                                    onclick="showAllDistances()">
269                                 Show Campus Distances
270                             </button>
271                         </td>
272                 </tr>
273
274                 <tr>
275                     <td colspan="2" align="center">
276                         <p class="note">
277                             Data Structures Project using Graphs
278                             and Dijkstra's Shortest Path Algorithm.
279                         </p>
280                     </td>
281                 </tr>
282
283             </table>
284         </form>
285
286   </body>
287   </html>
288
```

e/Desktop/logu/LOGU.HTML

Main Gate | Library | Canteen | CSE Block | Laboratory | Auditorium

Starting Location: Main Gate

Destination Location: CSE Block

Find Shortest Path

Shortest Path Result

Starting Point: Main Gate

Destination: CSE Block

Shortest Route: Main Gate → Library → CSE Block

Total Distance: 250 meters

Campus Connections

Distances between directly connected places

- Main Gate to Library : 100 meters
- Main Gate to Canteen : 200 meters
- Library to Canteen : 50 meters
- Library to CSE Block : 150 meters

Search

Find Shortest Path

Shortest Path Result

Starting Point: Main Gate

Destination: CSE Block

Shortest Route: Main Gate → Library → CSE Block

Total Distance: 250 meters

Campus Connections

Distances between directly connected places

- Main Gate to Library : 100 meters
- Main Gate to Canteen : 200 meters
- Library to Canteen : 50 meters
- Library to CSE Block : 150 meters
- Canteen to CSE Block : 100 meters
- Canteen to Laboratory : 200 meters
- CSE Block to Laboratory : 50 meters
- CSE Block to Auditorium : 120 meters
- Laboratory to Auditorium : 80 meters

Show Campus Distances

Data Structures Project using Graphs and Dijkstra's Shortest Path Algorithm.

COLLEGE CAMPUS SHORTEST PATH FINDER

ABSTRACT

The College Campus Shortest Path Finder is a data structures project designed to find the shortest and most efficient route between two locations within a college campus. A college campus may contain several buildings, classrooms, laboratories, libraries, hostels, canteens and administrative offices. Students, staff and visitors may sometimes find it difficult to identify the shortest route to reach a particular destination, especially when they are unfamiliar with the campus.

This project uses the Graph data structure and Dijkstra's shortest path algorithm to calculate the shortest distance between two selected locations. In this system, each campus location is represented as a vertex (node), and the paths connecting the locations are represented as edges. Each edge is assigned a weight based on the distance between two locations. The algorithm processes these distances and determines the minimum-distance route from the starting point to the destination.

The system accepts the source and destination locations as input and displays the shortest path along with the total distance. It helps users save time, reduce unnecessary walking, and navigate the campus more efficiently. The project can be implemented using the C programming language, with graphs and arrays used to represent locations and distances.

This project demonstrates the practical application of graph-based algorithms in solving real-world navigation problems. It also helps students understand important data structures and algorithm concepts through a useful campus-based application.

INTRODUCTION

Finding the shortest route between two places is a common problem in everyday life. In a large college campus, multiple paths may connect different buildings, and choosing the correct path can save both time and effort.

The College Campus Shortest Path Finder provides a simple solution to this problem. The user selects a starting location and a destination, and the system calculates the shortest available route using Dijkstra's algorithm. For example, a student who wants to travel from the college entrance to the library can use the system to identify the shortest route and its total distance.

The main purpose of this project is to apply the concepts of data structures and algorithms to a real-world problem. It provides an easy-to-understand method for representing campus locations and finding efficient routes between them.

COLLEGE CAMPUS SHORTEST PATH FINDER

OBJECTIVES

The main objectives of the College Campus Shortest Path Finder are:

1. To find the shortest route between any two selected campus locations.
2. To implement the Graph data structure for representing campus locations and connecting paths.
3. To apply Dijkstra's algorithm to calculate the minimum distance.
4. To reduce the time and effort required to navigate the campus.
5. To provide a simple and user-friendly navigation system.
6. To improve the practical understanding of data structures and algorithms.

WORKING PRINCIPLE

The system works through the following steps:

1. **Represent locations:** Each campus location is represented as a vertex in a graph.
2. **Represent paths:** The connections between locations are represented as edges.
3. **Assign distances:** Each edge is assigned a weight representing the distance between two locations.
4. **Select locations:** The user enters the starting point and destination.
5. **Apply the algorithm:** Dijkstra's algorithm calculates the shortest distance from the source to the destination.
6. **Display the result:** The system displays the shortest route and the total distance.

For example, if a user wants to travel from the main gate to the distary, the system checks the available routes and identifies the route with the minimum total distance.

ADVANTAGES

- Finds the shortest route between two locations.
- Saves time and reduces unnecessary walking.
- Helps new students and visitors navigate the campus.
- Provides practical implementation of graph concepts.
- Can be developed using the C programming language.
- Can be extended to support additional campus locations.

DISADVANTAGES

- The accuracy of the result depends on the correctness of the distance data.
- The basic system may not consider traffic, crowds or temporary road closures.
- New locations and paths must be added manually unless dynamic updates are implemented.
- Dijkstra's algorithm requires non-negative edge weights.
- A basic console-based implementation may not provide a graphical map.

CONCLUSION

The College Campus Shortest Path Finder is a useful application of graph data structures and Dijkstra's shortest path algorithm. It helps users identify the minimum-distance route between campus locations and demonstrates how computer science concepts can solve practical problems. In the future, the project can be enhanced with a graphical user interface, a campus map, live location tracking, and automatic updates to path information.